CSE 4107
Artificial Intelligence

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Fall 2021 Origin 42

Course Assignment

CSE4107 Total marks: 20

Fall 2021

Problems:

Assessment Mode: Viva on Report

Problems:

1. Explain the concept of Population in connection with Genetic algorithms. (4)

2. Perform an analysis through admissibility test of the heuristic function defined on a graph problem to decide whether it can be used as an instance of a type of informed search problems. (8)

3. Present a comparative study between Greedy Best-First Search and A* Search algorithms. (8)

Problems:

1. Explain the termination condition of a typical Genetic algorithm. (4)

2. Perform an analysis through consistency test of the heuristic function defined on a graph problem to decide whether it can be used as an instance of a type of informed search problems. (8)

3. Present a comparative study between Greedy Best-First Search and Hill-climbing Local Search algorithms. (8)

Problems:

1. Elaborate the concept of Beam Search. (4)

2. Analyze the role of Hill-climbing Local Search strategy in finding reasonable solutions to hard problems with no specific goal states. (8)

3. Present an evaluation of the A* search strategy. (8)

Problems:

1. Illustrate the concept of Local Maxima. (4)

2. Present an analysis targeting use of good Heuristic functions in solving real problems with huge search spaces. (8)

3. Evaluate the usefulness of biological themes in Genetic Algorithms for solving optimization problems. (8)

Problems:

1. Why is Greedy Best First search said to be incomplete? (4)

2. Present an analysis targeting negative aspects of the A* search strategy. (8)

3. Perform a comparative study between Hill-climbing local search and Genetic algorithms. (8)

Problems:

1. Elaborate the concept of State Space. (4)

2. Present an analysis with a view to avoiding local maxima in Hill-climbing local search. (8)

3. Evaluate the role of Greedy Best First search in solving certain hard problems. (8)

Class Test 2, Spring-20XX

Marks: 10 Time: 25 min.

1. How a planner exerts intelligence through domain feature consideration? (2)

2. Generate an optimal plan through Forward State Space search using PDDL for the planning problem in the Blocks world presented below. Omit the predicate ‘Block’, and give priority to gain in time over gain in actions. (3)

3. Why game playing problems are considered adversarial search problems? (2)

4. Construct a pruned game tree using Alpha-Beta pruning. Take the sequence, [6, 3, 2, 6, 4, 5, 7, 2, 8, 7, 5, 1, 3, 4] of MINIMAX values for the nodes at the cutoff depth of 4 plies. Assume that branching factor is 2, MAX makes the first move, and nodes are generated from left to right. (3)

1. How ‘precondition’ of actions are represented with PDDL? (2)

2. Generate an optimal plan using PDDL and Forward State Space search for the Air Cargo Transportation problem

given below. Omit predicates 'Airport', 'Cargo' and 'Plane', and consider generally accepted rules and notations for such a problem. (3)

A1	A2	A3
C1 C2	C3 C4	C5 C6
P1P2	P3 P4	P5 P6

Initial State

A1	A3
C3 C5	C4 C1

Goal State

3. Elaborate the concept of quiescence search. (2)

4. Construct a pruned game tree using Alpha-Beta pruning. Take the sequence, [1, 3, 4, 5, 7, 4, 6, 2, 1, 3, 5, 4, 8, 4, 6] of MINIMAX values for the nodes at the cutoff depth of 4 plies. Assume that branching factor is 2, MIN makes the first move, and nodes are generated from right to left. (3)

1. What role does utility theory play in acting under uncertainty? (2)
2. Consider the 'Escape the monster' environment given below.

OK 1,2 S	OK 2,2 S	
OK 1,1	OK 2,1	OK 3,1 S

Assume that OK indicates absence of a monster(M) at the cell, the status of the cells of the 4 x 4 grid other than those shown are unknown, and that the agent can smell(S) a monster at a cell vertically or horizontally adjacent to it.
Find $P(M_{2,3})$ taking 0.1 as the independent probability of a monster at any of the unknown cells. (3)

3. What are the major components of a Decision Network? (2)
4. Say, we have a Bayesian network containing 4 random variables A, B, C and D with two distinct values each. The conditional probability tables assigned to the random variables are given below. Draw the network and compute P (D | a $\wedge$ $\neg$b $\wedge$ c) using the network. (3)

A	B	P(c)
a	b	0.7
a	$\neg$b	0.5
$\neg$a	b	0.2
$\neg$a	$\neg$b	0.02

A	C	P(d)
a	c	0.8
a	$\neg$c	0.4
$\neg$a	c	0.3
$\neg$a	$\neg$c	0.01

B	P(a)
b	0.6
$\neg$b	0.05

P(b)
0.09

1. What are the major tools for acting under uncertainty? (2)

2. Joint-probability distribution of two random variables, having three different values each is shown in the table below.

		X_2		
		v_{21}	v_{22}	v_{23}
X_1	v_{11}	0.06	0.07	0.13
	v_{12}	0.21	0.14	0.12
	v_{13}	0.08	0.15	0.04

Compute the probabilities of the compound propositions, $v_{11} \vee v_{22}$ and $v_{21} | v_{13}$ using the distribution. (3)

3. Write a short note on 'Future state in Decision Networks'. (2)

4. Consider the 'Escape the monster' environment given below.

OK 1,2 S		
OK 1,2	OK 2,2 S	
OK 1,1	OK 2,1	OK 3,1 S

Assume that OK indicates absence of a monster(M) at the cell, the status of the cells of the 4 x 4 grid other than those shown are unknown, and that the agent can smell(S) a monster at a cell vertically or horizontally adjacent to it.
Find $P(M_{2,3})$ taking 0.3 as the independent probability of a monster at any of the unknown cells. (3)

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Class Test 1: Rational Agents, Knowledge Base & Reasoning.

Outline

1. Explain various aspects of the Agent concept with examples.
2. What are the characteristics of a rational agent?
3. Describe important concerns about the Environment for designing a rational agent.
4. Characterize and explain with examples Forward and Backward chaining algorithms with respect to inference in PL.
5. Give an illustrative example of limitations of logical inference.
6. Describe the syntactic rules of First Order Logic (FOL).
7. Express with an example the semantics of a query to knowledgebase in FOL.
8. How do you explain unification? Demonstrate the execution of the major steps of the simplified UNIFY algorithm using an example.
9. Explain with a suitable example the steps of conversion of a natural language sentence to a sentence in CNF of FOL.
10. State the resolution principle and demonstrate its application in proving the truth of a query to a KB in CNF of FOL.
11. Explain the resolution-refutation completeness of a KB.

Class Test 2

Course Outline

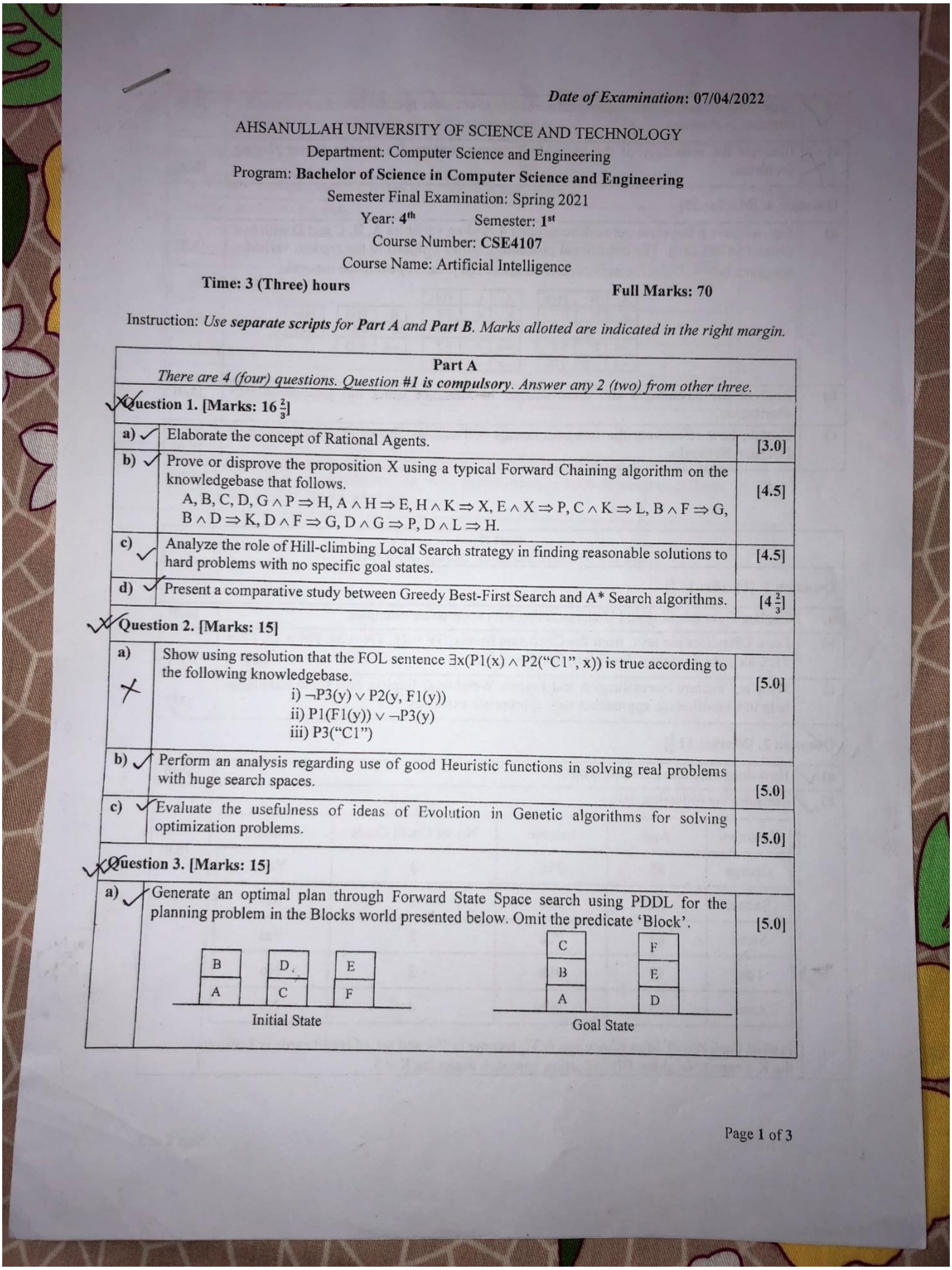
1. Describe the basic concepts of solving AI problems as search problems.
2. What do you mean by informed or heuristic search? Cite examples of heuristic functions.
3. Explain admissibility and consistency of heuristic functions with suitable examples.
4. Describe important characteristics of greedy best-first search.
5. Illustrate with an example how priority queues and possible paths are tracked in greedy best-first search.
6. Describe the evaluation function used in A* search.
7. Elaborate the important decisions about A* search and also the problems with it.
8. Illustrate with an appropriate example the execution of the A* search algorithm.
9. Explain the distinguishing features of hill-climbing local search with an example.
10. How the search environment in a hill-climbing local search is described using ‘state space landscape’, and what measures are taken to face the odds of the landscape?

11. What ideas stand behind genetic algorithms?

12. Describe the major steps of a typical genetic algorithm and illustrate them with examples.

13. Narrate important features of 'crossover' and 'mutation'.

Semester Final



Date of Examination: 07/04/2022

AHSANULLAH UNIVERSITY OF SCIENCE AND TECHNOLOGY
Department: Computer Science and Engineering
Program: **Bachelor of Science in Computer Science and Engineering**
Semester Final Examination: Spring 2021
Year: 4th Semester: 1st
Course Number: **CSE4107**
Course Name: Artificial Intelligence

Time: 3 (Three) hours **Full Marks: 70**

Instruction: *Use* ***separate scripts*** *for* ***Part A*** *and* ***Part B****. Marks allotted are indicated in the right margin.*

Part A

There are 4 (four) questions. Question #1 is ***compulsory****. Answer any 2 (two) from other three.*

Question 1. [Marks: $16\frac{2}{3}$]

a)	Elaborate the concept of Rational Agents.	[3.0]
b)	Prove or disprove the proposition X using a typical Forward Chaining algorithm on the knowledgebase that follows. $A, B, C, D, G \wedge P \Rightarrow H, A \wedge H \Rightarrow E, H \wedge K \Rightarrow X, E \wedge X \Rightarrow P, C \wedge K \Rightarrow L, B \wedge F \Rightarrow G, B \wedge D \Rightarrow K, D \wedge F \Rightarrow G, D \wedge G \Rightarrow P, D \wedge L \Rightarrow H.$	[4.5]
c)	Analyze the role of Hill-climbing Local Search strategy in finding reasonable solutions to hard problems with no specific goal states.	[4.5]
d)	Present a comparative study between Greedy Best-First Search and A* Search algorithms.	[$4\frac{2}{3}$]

Question 2. [Marks: 15]

a)	Show using resolution that the FOL sentence $\exists x(P1(x) \wedge P2(\text{"C1"}, x))$ is true according to the following knowledgebase. i) $\neg P3(y) \vee P2(y, F1(y))$ ii) $P1(F1(y)) \vee \neg P3(y)$ iii) $P3(\text{"C1"})$	[5.0]
b)	Perform an analysis regarding use of good Heuristic functions in solving real problems with huge search spaces.	[5.0]
c)	Evaluate the usefulness of ideas of Evolution in Genetic algorithms for solving optimization problems.	[5.0]

Question 3. [Marks: 15]

a)	Generate an optimal plan through Forward State Space search using PDDL for the planning problem in the Blocks world presented below. Omit the predicate 'Block'.	[5.0]

b)	Argue that the action schema is an efficient tool to consider specific features of diverse domains in Planners.	[5.0]
c)	Illustrate the suitability of the MINIMAX search strategy for solving Game Playing problems.	[5.0]

Question 4. [Marks: 15]

a)	Say, we have a Bayesian network containing 4 random variables A, B, C and D with two distinct values each. The conditional probability tables assigned to the random variables are given below. Draw the network and compute P(d | ¬a ∧ b) using the network.	[5.0]

A	B	P(c)
a	b	0.9
a	¬b	0.4
¬a	b	0.2
¬a	¬b	0.01

A	C	P(d)
a	c	0.8
a	¬c	0.5
¬a	c	0.3
¬a	¬c	0.02

B	P(a)
b	0.6
¬b	0.05

P(b)
0.07

b)	Analyze the advantages and disadvantages of inference using full joint-probability distribution.	[5.0]
c)	Explain how effectively the two components of Decision Theory are incorporated in Decision Networks.	[5.0]

Part B

There are 3 (three) questions. Answer any 2 (two).

Question 1. [Marks: $11\frac{2}{3}$]

a)	Explain Oversampling and Undersampling with appropriate examples.	[3.0]
b)	For a Classification task, from the Confusion Matrix, TP = 23, TN = 24, FN = 32, FP = 31. Calculate Accuracy, Precision, Recall and F Score.	[5.0]
c)	What are Feature Normalization and Feature Weighting? Explain how these techniques help in Classification approaches with appropriate examples.	[$3\frac{2}{3}$]

Question 2. [Marks: $11\frac{2}{3}$]

a)	How does Machine Learning work?	[3.0]
b)	Consider the following data.	[5.0]

Customer	Age	Income	No. of Credit Cards	Class
George	35	35k	3	Yes
Rachel	22	50k	1	No
Steve	63	70k	2	Yes
Tom	59	40k	2	No
Anne	25	29k	4	No

Predict the Class of John whose age is 37, income is 50k and no. of credit cards is 2 using the K Nearest Neighbor Classification approach assuming K = 3.

	What are Single Learner and Ensemble Learner approaches? Explain the significance of Ensemble Learners over Single Learners with appropriate examples.	[3 $\frac{2}{3}$]

Question 3. [Marks: 11 $\frac{2}{3}$]

a)	Explain how Backpropagation works in Artificial Neural Networks.	[3.0]
b)	Consider the following data. (see table below) Using K Means Clustering technique, partition the objects into two clusters.	[5.0]
c)	Consider a document containing 100 words wherein the word *tiger* appears 3 times. Also, there are 10 million documents, and the word *tiger* appears in 1000 documents of these. Calculate the TF-IDF for the word *tiger*.	[3 $\frac{2}{3}$]

Object	Attribute 1 (X): weight index	Attribute 2 (Y): pH value
Medicine A	1	1
Medicine B	2	1
Medicine C	4	3
Medicine D	5	4

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Semester final Part A

Question 1. [Marks: $13\,^{1}/_{3}$]

a) Think of an agent in the form of an autonomous vehicle that is supposed to room on the rocky surface of an alien planet and to collect data. Illustrate justified structural and functional components of it. **[6]**

b) Analyze the requirements of a knowledgebase for applying the chaining algorithms on it. Also, prove or disprove the proposition M using a typical forward chaining algorithm on the knowledgebase that follows. **[$2+5\,^{1}/_{3}$]**

A, B, C, D, $A \wedge B \Rightarrow R$, $A \wedge G \Rightarrow S$, $A \wedge P \Rightarrow E$, $B \wedge D \Rightarrow Q$, $B \wedge E \Rightarrow G$, $C \wedge Q \Rightarrow K$,
$D \wedge F \Rightarrow M$, $D \wedge K \Rightarrow P$, $D \wedge P \Rightarrow F$, $P \wedge R \Rightarrow H$.

Question 2. [Marks: $13\,^{1}/_{3}$]

a) Given in the following tables a weighted undirected graph along with the heuristic function values of the nodes, considering A as the initial node and G as the goal node. **[6]**

Node	A	A	B	B	B	C	C	D	D	E	E	F
Neighbor	B	C	C	D	E	E	F	E	G	F	G	G
Weight	46	54	47	36	25	35	32	68	62	57	58	45

Node	A	B	C	D	E	F	G
h(Node)	75	51	53	56	54	38	0

Find a solution to the problem instance demonstrating execution of the major steps of a typical Greedy Best-First search algorithm.

b) Illustrate the odds of the environment of an optimization problem containing various types of local maxima in it. Explain with it a situation when a randomly generated successor of a current state having a comparatively lesser value than the other successors will be more promising. **[$4+3\,^{1}/_{3}$]**

Question 3. [Marks: $13\,^{1}/_{3}$]

a) Why is Game Playing called an adversarial search problem? Explain the most important characteristics of a game playing problem with more than two players. **[$2+5\,^{1}/_{3}$]**

b) Consider a planning problem instance with the Air Cargo Transportation problem given below.

A1	A2	A3
C1 C2	C3 C4	C5 C6
P1 P2	P3 P4	P5 P6

Initial State

A1	A3
C4 C6	C3 C1

Goal State

Explain the process of executing some predefined actions for finding a solution to the instance. **[6]**

Question 4. [Marks: 13 $^{1}/_{3}$]

a) Full joint-probability distribution of two random variables, having three different values each is shown in the table below. The variables are assumed to describe the knowledgebase of a given domain uniquely. **[6]**

		X_2		
		v_{21}	v_{22}	v_{23}
X_1	v_{11}	0.05	0.08	0.13
	v_{12}	0.21	0.14	0.12
	v_{13}	0.09	0.15	0.03

Compute the probabilities of the compound propositions, $v_{12} \vee v_{23}$, $v_{12} | v_{21}$ and $v_{22} | v_{12}$ using the distribution. You need to apply the Bayes' rule to calculate one of the two conditional probabilities.

b) Write an illustrative essay on a possible way of application of the basic principle of decision theory. **[7 $^{1}/_{3}$]**

Part B

Instructions

i) The answer script **(Single PDF file for Part B)** must be uploaded at designated location in the provided **Google Form link** available in the Google classroom.

ii) Before uploading, rename the PDF file as follows.

CourseNo_StudentID_PartNo

Example: CSE4107_160204018_PartB.pdf

iii) There are **3 (Three)** questions in this part, answer **any 2 (Two).**

Table 1: Tennis Playing Dataset

Outlook	Temp	Windy	Play
sunny	hot	TRUE	no
sunny	hot	FALSE	yes
overcast	hot	FALSE	no
rainy	mild	FALSE	yes
sunny	cool	TRUE	yes
overcast	cool	TRUE	yes
overcast	cool	FALSE	yes
rainy	mild	FALSE	yes
rainy	cool	TRUE	no
rainy	mild	FALSE	no
overcast	mild	FALSE	yes
sunny	mild	FALSE	yes
sunny	hot	TRUE	yes

Table 2: Laptop Buying Dataset

age	student	income	rating	buy_laptop
26…35	yes	low	excellent	yes
<=25	no	medium	fair	no
<=25	yes	low	fair	yes
>35	yes	medium	fair	yes
<=25	yes	medium	excellent	yes
26…35	no	medium	excellent	yes
26…35	yes	high	fair	yes
<=25	no	high	fair	no
<=25	no	high	excellent	no
26…35	no	high	fair	yes
>35	no	medium	fair	yes
>35	yes	low	fair	yes
>35	yes	low	excellent	no
>35	no	medium	excellent	no

Question 1. [Marks: 10]		
a)	Consider the *Tennis Playing Dataset* (Table 1). Find the root node and draw the decision tree by following the Iterative Dichotomiser 3 algorithm.	[6]
b)	A binary dataset contains five different continuous features based on real-life scenario. Assume that (i) all the data are evenly distributed between two distinct classes and, (ii) all the data are of one class. Discuss the entropy and information gain based on these two assumptions.	[4]
Question 2. [Marks: 10]		
a)	For a multivariate dataset derive and minimize the sum of squared error equation in linear regression method. If the last two digits of your Student ID is characterized as '**A**', then find and sketch the best-fitting line equation for the data points (A, A+1), (A+1, A+2), (A+3, A+6), (A+4, A+4), (A+6, A+10).	[6]
b)	Discuss and find the difference among R-squared, MAE, MSE, and RMSE errors with an example. Which type of Naïve Bayes Model will efficiently work on categorical valued dataset and why?	[4]
Question 3. [Marks: 10]		
a)	Consider the *Laptop Buying Dataset* (Table 2). Find the class (*yes* or *no*) for an **X**-year-old student whose earning falls in *medium* category and who has an *excellent* rating. Here, X = last two digits of your student ID, but if X < 20, then X = last two digits of your ID + 20.	[6]

b)	Find the number of clusters and their features by applying the K-means clustering algorithm. The datapoints are characterized in Table 3 below. Also, to determine K in K-means clustering algorithm for working with unbiased data, which process will you adopt (Elbow or Silhouette) and why?	[**4**]

Table 3: Sample Dataset

		Datapoints	
		X_1	X_2
Features	A	5	3
	B	6	8
	C	3	8
	D	4	2
	E	7	0

AHSANULLAH UNIVERSITY OF SCIENCE AND TECHNOLOGY

Department: Computer Science and Engineering
Program: B.Sc. in Computer Science and Engineering
Semester Final Examination: Fall 2020
Year: 4th Semester: 1st
Course Number: CSE4107
Course Name: Artificial Intelligence

Time: 2 (Two) Hours **Full Marks: 50**

Semester Final

Use single answer script.

Instructions	i)	Answer scripts should be handwritten and should be written in A4 white paper. You must submit the hard copy of the answer script to the Department when the universityreopens.
	ii)	You must write the following information on the top page of your answer script: Department: Program: Course no.: Course Name: Examination: Semester (Session): Student ID: Signature and Date:
	iii)	Write down Student ID, Course number and put your signature on top of every single page of the answer script.
	iv)	Write down page number at the bottom of every page of the answer script.
	v)	Upload the scan copy of your answer script in PDF format through provided **google form** at the respective course site (i.e., **google classroom)** using institutional email within the allocated time. Uploading clear and readable scan copy (uncorrupted) is your responsibility and you must cover all the pages of your answer script. However, for clear and readable scan copy of the answer script student should use only one side of a page for answering the questions.
	vi)	You must avoid **plagiarism** and, maintain **academic integrity and ethics**. You are not allowed to take any help from another individual. If you do so, it can result in stern disciplinary actionsagainst you from the university authority.
	vii)	**Marks** allotted are indicated in the **right margin.**
	viii)	Assume any reasonable data if needed.
	ix)	Symbols and characters have their usual meaning.
	x)	Before uploading, rename the PDF file as follows. **CourseNo_StudentID** Example: CSE4107_170204025.pdf

There are **6 (Six)** questions, answer **any 4 (Four)**.

Question 1. [Marks: 12.5]

a) Derive a most general unifier for the following set of FOL sentences. **[4]**

S = {P_i(x, F1(Toy), y, u), P_i(Happy, z, F2(x), v)}.

b) Analyze the role of issues related to environment of functioning in designing rationalagents. **[4.5]**

c) Present a comparative study of the forward chaining and backward chaining algorithms. **[4]**

Question 2. [Marks: 12.5]

a) Given below through the tables is a weighted undirected graph along with the heuristicfunction values of the nodes. [4]

Show that the heuristic function is consistent.

Node	A	A	B	B	B	C	C	D	D	E	F
Neighbor	B	C	C	D	E	D	F	E	G	G	G
Weight	45	42	37	36	28	26	32	24	47	36	41

Node	A	B	C	D	E	F	G
h(Node)	70	50	52	38	33	30	0

b) Present an analysis of the biological inspiration deployed in common Genetic algorithms. [4.5]

c) Demonstrate a comparison between A* search and Hill-climbing local search. [4]

Question 3. [Marks: 12.5]

a) Construct a pruned game tree using Alpha-Beta pruning. Take the sequence, [5, 3, 2, 4, 1, 3, 6, 2, 7, 8, 5, 4, 1, 6] of MINIMAX values for the nodes at the cutoff depth of 4 plies and assume that branching factor is 2, MIN makes the first move, and nodes are generated from *right to left*. [4]

b) Analyze the characteristics of a planner to establish that it is an intelligent agent. [4.5]

c) Compare the features of search strategies used for planners and for game playing algorithms. [4]

Question 4. [Marks: 12.5]

a) Say, we have a Bayesian network containing 4 random variables A, B, C and D with two distinct values each. The conditional probability tables assigned to the random variables are given below. Draw the network and compute P(C | a $\wedge$ ¬b $\wedge$ ¬d) using the network. [4]

A	B	P(d)
a	b	0.9
a	¬b	0.6
¬a	b	0.3
¬a	¬b	0.01

A	D	P(c)
a	d	0.8
a	¬d	0.4
¬a	d	0.2
¬a	¬d	0.02

B	P(a)
b	0.7
¬b	0.04

P(b)
0.08

b)	Analyze the evaluation steps of a Decision network to justify that the basic principle ofdecision theory underlies the whole process.	[**4.5**]
c)	Briefly demonstrate a comparison between inference using full joint probability distributionand inference using Bayes' theorem.	[**4**]

Question 5. [Marks: 12.5]

a)	Consider the 'Escape the monster' environment given below.	[**4**]

OK 1,2	OK 2,2 S	
OK 1,1	OK 2,1	OK 3,1 S

Assume that OK indicates absence of a monster(M) atthe cell, the status of the cells of the 4 x 4 grid other than those shown are unknown, and that the agent can smell(S) a monster at a cell vertically or horizontally adjacent to it.

Find $P(M_{4,1})$ taking 0.3 as the independent probability of a monster at any of the unknown cells.

b)	How do you interpret the final output of a typical Decision Tree learning algorithm?	[**4.5**]
c)	Evaluate the advantages and disadvantages of Neural Network learning with respect toother machine learning algorithms.	[**4**]

Question 6. [Marks: 12.5]

a)	Given below is a table that describes some 3-attribute training samples with their class tags.	[**4**]

A1	A2	A3	Class
1	H	a	N
1	L	a	P
1	H	b	N
2	H	a	P
2	L	a	N
2	H	b	N

Find a Naïve Bayes prediction for the sample, (1, L, b).

b)	How do you characterize the application domains of clustering algorithms?	[**4.5**]
c)	Present a brief comparison between the use of k-Nearest Neighbor classifiers and the useof Support Vector machines as classifiers.	[**4**]

Department: Computer Science and Engineering
Program: B.Sc. in Computer Science and Engineering
Semester Carryover/Clearance/Improvement Examination: Fall 2020
Year: 4th Semester: 1st
Course Number: CSE4107
Course Name: Artificial Intelligence

Time: 2 (Three) Hours **Full Marks: 50**

Carryover/Clearance/Improvement Examination

Use single Answer Script.

Instructions	i)	Answer scripts should be handwritten and should be written in A4 white paper. You must submit the hard copy of the answer script to the Department when the universityreopens.
	ii)	You must write the following information at the **top page of each part**: Department: Program: Course no.: Course Title: Examination: Semester (Session): Student ID: Signature and Date:
	iii)	Write down Student ID, Course number and put your signature on top of every single page ofthe answer script.
	iv)	Write down the page number at the bottom of every page of the answer script.
	v)	Upload the scanned copy of your answer script in PDF format through provided **Google Form** link within the allocated time. Uploading clear and readable scan copy (uncorrupted) is your responsibility and you must cover all the pages of your answer script. However, for clear and readable scanned copy of the answer script a student should use only one side of a page for answering the questions.
	vi)	You must avoid **plagiarism** and, maintain **academic integrity and ethics**. You are not allowedto take any help from another individual. If you do so, it can result in stern disciplinary actions against you from the university authority.
	vii)	**Marks** allotted are indicated in the **right margin.**
	viii)	Assume any reasonable data if needed.
	ix)	Symbols and characters have their usual meaning.
	x)	Before uploading the answer script **(single PDF file)**, rename the file as follows. **CourseNo_StudentID.pdf** Example: CSE4107_160204018.pdf

Question 1. [Marks: 12.5]

a) Prove or disprove the proposition H using a typical backward chaining algorithm on theknowledgebase that follows. **[4]**

A, B, C, D, $A \wedge G \Rightarrow P$, $A \wedge P \Rightarrow E$, $C \wedge Q \Rightarrow M$, $P \wedge R \Rightarrow H$, $B \wedge E \Rightarrow G$, $B \wedge D \Rightarrow Q$, $D \wedge F \Rightarrow G$, $D \wedge P \Rightarrow F$, $D \wedge M \Rightarrow P$, $A \wedge B \Rightarrow R$.

b) Analyze the role of proof by contradiction in finding answer to queries to knowledgebasesrepresented in First Order Logic (FOL) through resolution. **[4.5]**

c) Present a comparative study between top-down and bottom-up approaches in syntactic analysisof FOL sentences. **[4]**

Question 2. [Marks: 12.5]

a) Given below through the tables is a weighted undirected graph along with the heuristic functionvalues of the nodes. **[4]**

Node	A	A	B	B	B	C	C	D	D	E	E	F
Neighbor	B	C	C	D	E	E	F	E	G	F	G	G
Weight	46	54	47	36	25	35	32	68	62	57	58	45

Node	A	B	C	D	E	F	G
h(Node)	75	51	53	56	54	38	0

Demonstrate the process of finding a solution to the problem instance taking A as the initialnode and using Greedy Best First search.

b) Present an analysis of positive and negative sides of local search algorithms. **[4.5]**

c) Compare the ideas that underly Best-First graph search and Stochastic Beam search algorithms. **[4]**

Question 3. [Marks: 12.5]

a) Generate an optimal plan through Forward State Space search using PDDL for the planning problem in the Blocks world presented below. Omit the predicate 'Block' and give priority to gain in number of steps over gain in number of actions. **[4]**

				A	D
B	D	F		B	E
A	C	E		C	F

Initial State

Goal State

b) Analyze the characteristics of the MINIMAX search strategy applied in game playing. **[4.5]**

c) Compare the features of search spaces used for planners and for game playing algorithms. **[4]**

Question 4. [Marks: 12.5]

a) Full joint-probability distribution of two random variables having three different values eachis shown in the table below. **[4]**

		X_2		
		v_{21}	v_{22}	v_{23}
X_1	v_{11}	0.06	0.07	0.13
	v_{12}	0.21	0.14	0.12
	v_{13}	0.08	0.15	0.04

Compute the probabilities of the compound propositions, $v_{13}|v_{23}$ and $v_{22}|v_{11} \vee v_{12}$ using the distribution.

b) Present an analysis of the purpose of Bayesian Networks. **[4.5]**

c) Compare the role of chance nodes and decision nodes in influencing the outcome of Decision Networks. **[4]**

Question 5. [Marks: 12.5]

a) Consider the 'Escape the monster' environment given below. **[4]**

OK 1,2	OK 2,2 S	
OK 1,1	OK 2,1	OK 3,1 S

Assume that OK indicates absence of a monster(M) atthe cell, the status of the cells of the 64 x 64 grid otherthan those shown are unknown, and that the agent can smell(S) a monster at a cell vertically or horizontally adjacent to it.

Find $P(M_{4,1})$ taking 0.1 as the independent probability of a monster at any of the unknown cells.

b) How do you explain the process of learning that occurs in neural networks functioning based on the backpropagation algorithm? **[4.5]**

c) Present a brief comparison between the supervised and unsupervised machine learning strategies. **[4]**

Question 6. [Marks: 12.5]

a) Given below is a table that describes some 3-attribute training samples with their class tags. **[4]**

A1	A2	A3	Class
1	H	a	N
1	L	a	P
1	H	b	N
2	H	a	P
2	L	a	N
2	H	b	N

Find the root of the Decision Tree satisfying the samples.

b) Present an analysis of the idea behind k-Nearest Neighbor classifiers. **[4.5]**

c) How do you evaluate the importance of clustering algorithms? **[4]**

Spring 2019

Quiz 01

CSE4107
Class Test 1, Spring-2019

Marks: 10 Time: 25 min.

1. What are the basic components of an Agent? (2)
2. Prove or disprove the proposition P using a typical forward chaining algorithm on the knowledgebase that follows. (3)

$A, B, C, D, A \wedge G \Rightarrow H, A \wedge H \Rightarrow E, C \wedge K \Rightarrow L, H \wedge K \Rightarrow P, B \wedge E \Rightarrow G, B \wedge D \Rightarrow K,$
$D \wedge F \Rightarrow G, D \wedge H \Rightarrow F, D \wedge L \Rightarrow H.$

3. How a Term is defined in the syntax of First Order Logic? (2)
4. Illustrate with an example the major steps of a typical unification algorithm. (3)

------

1. When an Agent is said to be rational? (2)
2. Prove or disprove the proposition H using a typical backward chaining algorithm on the knowledgebase that follows. (3)

$A, B, C, D, A \wedge G \Rightarrow H, A \wedge H \Rightarrow E, C \wedge K \Rightarrow L, H \wedge K \Rightarrow P, B \wedge E \Rightarrow G, B \wedge D \Rightarrow K,$
$D \wedge F \Rightarrow G, D \wedge H \Rightarrow F, D \wedge L \Rightarrow H.$

3. How do you define the syntax and semantics of First Order Logic? (2)
4. Describe the major steps of proving a natural language sentence using resolution on a knowledgebase. (3)

----

1. What are the main issues in specification of task environment for designing a rational agent? (2)
2. Illustrate with an example the fact that forward chaining algorithms are data driven. (3)
3. Explain the matching rules that are involved during Unification. (2)
4. Convert the following FOL sentence into CNF. (3)

$\forall x\, (Student(x) \wedge Takes(x, CSE4107) \leftrightarrow CSE\_Student(x)).$

----

1. What are the basic types of Agents? (2)
2. Explain with an example that backward chaining algorithms are said to be goal oriented. (3)
3. Show that $\exists x(Student(x) \wedge Plays(x, Cricket))$ is a valid sentence of First Order Logic. (2)
4. Derive P2(Best, Rest) from the following clauses using the Resolution rule systematically. (3)

   $P1(x, F1(y)) \vee P2(x, u), \neg P1(Best, z) \vee P3(u, F2(x)), \neg P3(Rest, v).$

----

1. How do you explain the concept of Percepts and Actions of an Agent? (2)
2. Explain the characteristics of forward chaining algorithms. (3)
3. What are the major steps of conversion of FOL sentences to CNF? (2)
4. Derive a most general unifier for the following set of sentences. (3)

   $S = \{P2(x, F1(Mia), Shaheb, y), P2(Babu, v, u, F2(x))\}.$

----

1. What contrasting features of environment are generally considered for designing rational agents? (2)
2. Explain the characteristics of backward chaining algorithms. (3)
3. How do you explain the concept of unification? (2)
4. Show that

$\forall x, y, z\ ((Brother(x, y) \wedge Son(z, x)\ ) \Rightarrow Nephew(z, y))$ is a syntactically correct sentence of First Order Logic. (3)

1. Write a brief illustrative essay on Actions and Actuators of an Agent. (3)
2. Prove or disprove the proposition S using a typical backward chaining algorithm on the knowledgebase that follows. (4)

$A, B, C, D, A \wedge G \Rightarrow P, A \wedge P \Rightarrow E, C \wedge Q \Rightarrow M, P \wedge R \Rightarrow S, B \wedge E \Rightarrow G, B \wedge D \Rightarrow Q,$
$D \wedge F \Rightarrow G, D \wedge P \Rightarrow F, D \wedge M \Rightarrow P, A \wedge B \Rightarrow R.$

3. How do you explain the concept of Unification? (3)

----

1. Write a brief illustrative essay on Sensors and Percepts of an Agent. (3)
2. Demonstrate that

$\forall x, y, z\ ((Father(z, x) \wedge Father(z, y) \wedge Male(x) \wedge \neg(x = y)) \Rightarrow brother(x, y))$ is a syntactically correct FOL sentence. (4)

3. How Resolution is used to prove or disprove a sentence? (3)

----

1. Discuss the Environment issues that are considered for designing rational agents. (3)
2. Convert the following English sentence to CNF of FOL. (4).

    A student of CSE department must take CSE1101 and CSE4107.

3. Distinguish between forward and backward chaining algorithms. (3)

----

1. Elaborate the facts that make an Agent rational. (3)
2. Prove or disprove $\exists x(Math(x) \wedge Takes\ (Karim, x))$ from the following knowledgebase using resolution. (4)
    a) Student(Karim)
    b) $Math(MathFor(y)) \vee \neg Student(y)$
    c) $\neg Student(y) \vee Takes(y, MathFor(y))$
3. Describe a situation when inferences based on pure logic fails in choosing from alternatives. (3)

----

1. How do you characterize the knowledgebase for reasoning with propositional logic? (3)
2. What is an mgu (most general unifier)? Derive an mgu, if possible, for the following set of FOL sentences. (1+3)

S = {P1(F2(Miki), z, Blue, u), P1(y, Red, x, F3(z))}.

3. Explain with an illustrative example the semantics of FOL. (3)

----

1. Explain the characteristics of forward chaining algorithms. (3)
2. Take the following situation of 'Find Gold avoiding Monsters' problem, and derive $\neg M_{3,2}$. (4)

$S_{i,j} \Leftrightarrow M_{i-1,j} \vee M_{i+1,j} \vee M_{i,j-1} \vee M_{i,j+1}$ $\qquad (i \geq 1) \wedge (j \geq 1) \wedge (i \leq 4) \wedge (j \leq 4)$

$\neg M_{3,1} \wedge \neg S_{3,1} \wedge \neg G_{3,1}$ $\qquad A_{3,1}$

3. How Atomic Sentences are defined in the syntax of FOL? (3)

1. Discuss about possible 'Sensors' and 'Actuators' of an automatic car. (3)
2. Prove or disprove the proposition P using a typical forward chaining algorithm on the knowledgebase that follows. (4)

$A, B, C, D, G \wedge P \Rightarrow H, A \wedge H \Rightarrow E, C \wedge K \Rightarrow L, H \wedge K \Rightarrow S, B \wedge F \Rightarrow G, B \wedge D \Rightarrow K,$
$D \wedge F \Rightarrow G, D \wedge G \Rightarrow P, D \wedge L \Rightarrow H.$

3. Explain resolution-refutation completeness of a knowledgebase. (3)

1. How do you explain that rational Agents do the 'right' thing? (3)
2. Demonstrate that

$\forall x, y, z\ ((\text{Mother}(z, x) \wedge \text{Mother}(z, y) \wedge \text{Female}(x) \wedge \neg(x = y)) \Rightarrow \text{Sister}(x, y))$

is a syntactically correct FOL sentence. (4)

3. Write a brief essay on Unification. (3)

----

1. What are the basic types of agents? (2)
2. Why backward chaining algorithms are said to be goal oriented? (2)
3. Explain the meaning of semantics of FOL. (2)
4. Prove or disprove $\exists x(\text{Course}(x) \wedge \text{Takes}(\text{Hasan}, x))$ from the following knowledgebase using resolution. (4)

b) Student(Hasan) b) Course(CourseFor(x)) $\vee$ ¬Student(x) c) ¬Student(x) $\vee$ Takes(x,CourseFor(x))

----

1. Explain the concept of Percepts of an Agent. (2)
2. Illustrate the fact that an agent making inferences based on pure logic may fail in choosing from alternatives evenin a simple situation. (3)
3. What are the matching criteria for Unification? (2)
4. Convert the following English sentence to CNF of FOL. (3) Every student must take at least one course of HUM type.

------

1. Why learning capability is a must for a rational Agent? (2)
2. Illustrate with an example how resolution is used to prove or disprove sentences? (3)
3. Demonstrate that

$\forall x, y, z\ ((\text{Brother}(x, y) \wedge \text{Son}(z, x)) \Rightarrow \text{Nephew}(z, y))$

is a syntactically correct FOL sentence. (3)

4. How do you characterize forward chaining algorithms? (2)

------

1. Explain the differences between forward and backward chaining algorithms. (3)
2. Consider the following situation of 'Find Gold avoiding Monsters' problem.

$S_{i,j} \Leftrightarrow M_{i-1,j} \vee M_{i+1,j} \vee M_{i,j-1} \vee M_{i,j+1}$ $(i \geq 1) \wedge (j \geq 1) \wedge (i \leq 4) \wedge (j \leq 4)$

$\neg M_{1,1} \wedge \neg M_{2,1} \wedge S_{2,1} \wedge \neg G_{2,1}$ $A_{2,1}$

Derive the fact that $M_{3,1} \vee M_{2,2}$ using inference rules of propositional logic. (4)

3. Describe the major steps of conversion of FOL sentences to CNF. (3)

Quiz 02

CSE4107
Class Test 2, Spring 2019
Marks: 10 Time: 25 min.

1. Write a short note on 'Successor function'. (2)
2. What are the main differences between A* search and Greedy Best-First search? (3)
3. Explain the positive side of Hill-climbing local search. (2)
4. Elaborate the concept of New Population in connection with Genetic algorithms. (3)

1. Write a short note on 'State Space'. (2)
2. Illustrate the concept of global maximum in connection with Hill-climbing local search. (3)
3. What do you know about Priority Queues? (2)
4. Explain the importance of Crossover used in Genetic algorithms. (3)

1. Explain the positive side of Greedy Best-First search. (3)
2. Justify the naming of the strategy called Hill-climbing Local Search. (3)
3. How the parent generation is formed in Genetic algorithms? (2)
4. Explain the positive side of Greedy Best-First search. (2)

1. Illustrate the concept of consistency of heuristic functions with a concise example. (3)
2. Why a genetic algorithm is called a stochastic beam search? (2)
3. Explain the positive and negative sides of A* search. (3)
4. Write a short note on First-choice Hill-climbing search. (2)

1. Write a short note on 'Local search'. (2)
2. Explain the major steps of A* search. (3)
3. Write a brief essay on heuristic functions. (3)
4. How evolution is modeled in Genetic algorithms? (2)

1. Distinguish between blind and informed search. (2)
2. Describe the major steps of a typical Hill-climbing local search algorithm. (3)
3. How reproduction is modeled in Genetic algorithms? (2)
4. Illustrate the process of updating the Possible Path in Greedy Best-first search. (3)

-

1. How do you explain a Goal test in search problems? (2)

2. Explain three important decisions about A* search. (3)
3. Write a brief essay on 'Some real problems require Hill-climbing local search'. (3)
4. Elaborate the concept of Mutation in connection with Genetic algorithms. (2)

1. Write a short note on 'Optimal solution of a search problem'. (2)
2. Illustrate the odds of the environment of Hill-climbing local search problems. (3)
3. What do you know about the evaluation function of Greedy best-first search? (2)
4. Narrate the ideas that stand behind Genetic algorithms. (3)

1. Describe the conditions under which heuristic functions are considered good. (3)
2. Explain three of the most important features of Hill-climbing Local Search. (3)
3. How the new generation is formed in Genetic algorithms? (2)
4. Mark the negative points in Greedy Best-First search. (2)

1. Explain the necessity of heuristic functions for problem solving. (2)
2. Narrate those major steps of a typical genetic algorithm that are repeatedly executed. (3)
3. What are the problems with A* search, and how those can be addressed? (3)
4. Write a short note on Random restart Hill-climbing search. (2)

1. Write a short note on 'Local maximum'. (2)
2. Narrate the major steps of A* search that are repeatedly executed. (3)
3. Write a brief essay on Consistency of heuristic functions. (3)
4. How Crossover is executed in Genetic algorithms? (2)

1. Write a short note on 'Admissibility of heuristic functions'. (2)
2. Describe the repeatedly executed major steps of a typical Hill-climbing local search. (3)
3. How randomness is applied in Genetic algorithms? (2)
4. Explain the positive and negative sides of Greedy Best-first search. (3)

1. Write a short note on 'Solution to AI problems'. (2)
2. Why and how a tree of nodes is maintained in Greedy Best-first search? (3)
3. For what type of problems Hill-climbing local search is appropriate? (2)
4. Elaborate the concept of Population in connection with Genetic algorithms. (3)

1. Write a short note on 'Path cost of AI problems'. (2)
2. Illustrate the concept of local maxima in connection with Hill-climbing local search. (3)
3. What do you know about the evaluation function of A* search? (2)

4. Narrate the importance of Crossover and Mutation used in Genetic algorithms. (3)

1. Illustrate the conditions under which heuristic functions are considered good. (3)
2. Justify the naming of the strategy called Hill-climbing Local Search. (3)
3. How the new generation is formed in Genetic algorithms? (2)
4. Explain the negative side of A* search. (2)

1. Illustrate the concept of consistency of heuristic functions with a concise example. (3)
2. Why a genetic algorithm is called a local search? (2)
3. Explain the positive and negative sides of Greedy Best-first search. (3)
4. Write a short note on 'Stochastic Hill-climbing search'. (2)

1. Write a short note on 'Hill-climbing search'. (2)
2. Narrate the major steps of A* search that are repeatedly executed. (3)
3. Illustrate the concept of heuristic functions. (3)
4. How Crossover is executed in Genetic algorithms? (2)

1. Distinguish between blind and informed search. (2)
2. Describe the repeatedly executed major steps of a typical Genetic algorithm. (3)
3. Explain the positive side of Hill-climbing local search. (2)
4. Illustrate the process of updating the Possible Path in A* search. (3)

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Class Test 2, Spring 2019
Marks: 10 Time: 25 min.

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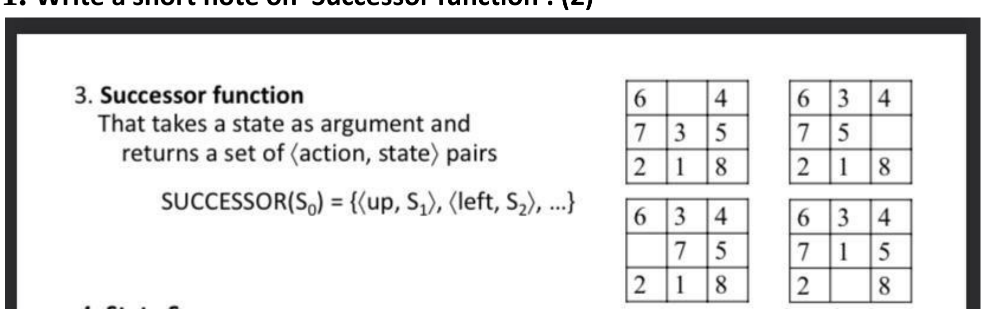
1. Write a short note on 'Successor function'. (2)



2. What are the main differences between A* search and Greedy Best-First search? (3)
Ans:

8

+50

Both algorithms fall into the category of "best-first search" algorithms, which are algorithms that *can* use both the knowledge acquired so far while exploring the search space, denoted by $g(n)$, and a heuristic function, denoted by $h(n)$, which estimates the distance to the goal node, for each node n in the search space (often represented as a graph).

Each of these search algorithms defines an "evaluation function", for each node n in the graph (or search space), denoted by $f(n)$. This evaluation function is used to determine which node, while searching, is "expanded" first, that is, which node is first removed from the "fringe" (or "frontier", or "border"), so as to "visit" its children. In general, the difference between the algorithms in the "best-first" category is in the definition of the evaluation function $f(n)$.

11:19
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In the case of *the greedy BFS algorithm*, the evaluation function is $f(n) = h(n)$, that is, the greedy BFS algorithm first expands the node whose estimated distance to the goal is the smallest. So, greedy BFS does not use the "past knowledge", i.e. $g(n)$. Hence its connotation "greedy". In general, the greedy BST algorithm is **not complete**, that is, there is always the risk to take a path that does not bring to the goal. In the greedy BFS algorithm, all nodes on the *border* (or fringe or frontier) are kept in memory, and nodes that have already been expanded do not need to be stored in memory and can therefore be discarded. In general, the greedy BFS is also **not optimal**, that is, the path found may not be the optimal one. In general, the time complexity is $\mathcal{O}(b^m)$, where b is the (maximum) branching factor and m is the maximum depth of the search tree. The space complexity is proportional to the number of nodes in the fringe and to the length of the found path.

3. Explain the positive side of Hill-climbing local search. (2) Ans:

C) Distinguishing features of Hill-climbing Local Search

1. Uses a **single current state**;
2. **Investigates neighbors** generated by the successor function;
3. Terminates if a state with optimal / 'sufficient' value found;
4. Generally does not involve a goal state;
5. Like **greedy best-first**; Selects from neighbors the one that appears best according to a **heuristic function**, that is, that appears to lead most quickly to the 'top of the hill';
6. Requires little memory;
7. This may work better than those that consider paths, for very large or potentially infinite search spaces.

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In each case, the algorithm reaches a point at which no progress is being made. Starting from a randomly generated 8-queens state, steepest-ascent hill climbing gets stuck 86% of the time, solving only 14% of problem instances. It works quickly, taking just 4 steps on average when it succeeds and 3 when it gets stuck—not bad for a state space with 88 ≈ 17 million states.

4. Elaborate the concept of New Population in connection with Genetic algorithms. (3)

Ans: For example, the first child of the first pair gets the first three digits from the first parent and the remaining digits from the second parent, whereas the second child gets the first three digits from the second parent and the rest from the first parent. The 8-queens states involved in this reproduction step are shown in Figure 4.7. The example shows that when two parent states are quite different, the crossover operation can produce a state that is a long way from either parent state. It is often the case that the population is quite diverse early on in the process, so crossover (like simulated annealing) frequently takes large steps in the state space early in the search process and smaller steps later on when most individuals are quite similar.

3.5. Genetic Algorithms

A) Ideas Behind

1) A GA is a variant of stochastic beam search:

- local search
- more than one current states
- probabilistic in choosing the successor state

2) A GA models reproduction involving two parents:
a successor is generated by combining two states

3) A GA models evolution:

- population genetics
- survival of the fittest

✓Parent states are chosen from a set after rating with an objective function.
✓The function is called fitness function .
✓ 'Crossover' and 'Mutation' are also accommodated.

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1. Write a short note on 'State Space'. (2) Ans:

	7	5
2	1	8

7	1	5
2		8

4. **State Space**
Set of states reachable from the initial state;
Initial state + successor function

$S_0, S_1, S_2, \ldots$
S_0, SUCCESSOR

5. **Path**

2. Illustrate the concept of global maximum in connection with Hill-climbing local search. (3) Ans:

Hill Climbing is a heuristic search used for mathematical optimization problems in the field of Artificial Intelligence.
Given a large set of inputs and a good heuristic function, it tries to find a sufficiently good solution to the problem. This solution may not be the global optimal maximum.

2. **Global maximum:** It is the best possible state in the state space diagram. This is because, at this stage, the objective function has the highest value.

3. What do you know about Priority Queues? (2) Ans:

Priority Queue is an abstract data type, which is similar to a queue, however, in the

Without crossover, all you have is local mutations. This means change will happen slowly, and it will be very hard to get your population out of a local optimum.

With crossover, you can combine partial solutions from different candidates. This often means making a pretty huge jump from either of the parents, which means you can move out of a local optimum a bit more easily.

For crossover to work, you need to make sure that your representation makes it possible for crossover to recombine into something sensible.

4. Explain the importance of Crossover used in Genetic algorithms. (3) Ans:

Crossover is the mechanism that lets a GA share information about different parts of the solution space between different candidate solutions.

If you removed crossover, you're left with just mutation, and the algorithm essentially becomes randomised local search, done in parallel for each starting candidate solution.

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1. **Explain the positive side of Greedy Best-First search. (3)**

Ans: Greedy best-first search algorithm always selects the path which appears best at that moment. It is the combination of depth-first search and breadth-first search algorithms. It uses the heuristic function and search. Best-first search allows us to take the advantages of both algorithms. With the help of best-first search, at each step, we can choose the most promising node. In the best first search algorithm, we expand the node which is closest to the goal node and the closest cost is estimated by a heuristic function,f(n)=g(n). A priority queue is maintained that contains nodes in ascending order of h-values.

2. **Justify the naming of the strategy called Hill-Climbing Local Search? (2)**

Ans: The hill-climbing search algorithm is the most basic local search technique. Hill climbing algorithm continuously moves in the direction of increasing elevation/value to find the peak of the mountain or best solution to the problem. It terminates when it reaches a peak value where no neighbor has a higher value. Hill climbing is sometimes called greedy local search because it grabs a good neighbor state without thinking ahead about where to go next. Although greed is considered one of the seven deadly sins, it turns out that greedy algorithms often perform quite well. So, the naming strategy of hill climbing local search is justified.

3. **How the parent generation is formed in Genetic algorithms? (2)**

Ans: In genetic algorithms, parent generation is formed by taking better performing individuals from the fitness functions. Suppose we have 3 individuals rated using a fitness function where F(a)=23, F(b)=13, F(c)=22. Now, if 15 is the threshold value, parent generation is {a,c}. This is how parent generation is formed by a genetic algorithm.

4. **Explain the positive side of Greedy Best-First search. (2) Ans:**

- Can switch between BFS and DFS, thus gaining the advantages of both.
- More efficient when compared to DFS.
- Node is expanded which is closest to the goal node and the closest cost is estimated by a heuristic function, f(n)=g(n).

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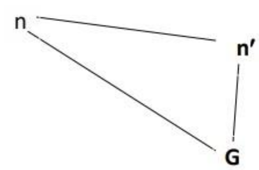
1. Illustrate the concept of consistency of heuristic functions with a concise example. (3)

Ans:

g) Monotonicity or Consistency of heuristic functions

$h(n) \leq c(n, a, n') + h(n')$, for any node n and every successor n′ of n

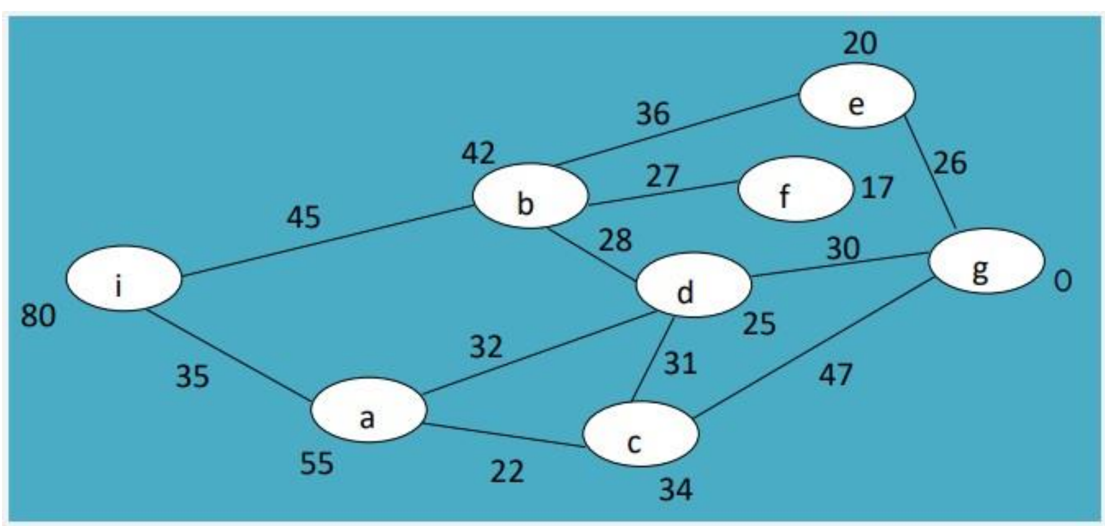
It is from triangular inequality: one side is shorter than the sum of the other two.



❖ h_1 and h_2 seen above are also consistent.

❖ Every consistent heuristic function is also admissible, but vice-versa is not always true.

[May work very well on n′, while not so well on n.]



h(n) ≤ c(n, a, n\`) + h(n\`), for any node n and every successor n\` of n h(i)=80 c(i, →, b) + h(b)=45+42, h(i)=80 c(i, →, a) + h(a)=35+55;

2. Why is a genetic algorithm called a stochastic beam search? (2)

Ans: Stochastic Beam Search (SBS) is a simple search procedure that starts with a set of initial solutions and explores their neighborhood solutions using successor functions. It then keeps track of only the most promising (in terms of fitness) k solutions.

A genetic algorithm (or GA) is a variant of stochastic beam search in which successor states are generated by combining two parent states rather than by modifying a single state. The analogy to natural selection is the same as in stochastic beam search, except that now we are dealing with sexual rather than asexual reproduction.

3. Explain the positive and negative sides of A* search. (3) Ans:

Positive:

- It is an optimal search algorithm in terms of heuristics.
- It is one of the best heuristic search techniques.
- It is used to solve complex search problems.
- There is no other optimal algorithm guaranteed to expand fewer nodes than A*.

Negative:

- This algorithm is complete if the branching factor is finite and every action has fixed cost.
- The performance of A* search is dependent on accuracy of the heuristic algorithm used to compute the function h(n).

4. Write a short note on First-choice Hill-climbing search. (2)

Ans: First-choice hill climbing implements stochastic hill climbing by generating successors randomly until one is generated that is better than the current state. This is a good strategy when a state has many (e.g., thousands) of successors.

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1. Write a short note on 'Local search'. (2)

Ans: Local search algorithms operate using a single current node rather than multiple paths and generally move only to neighbors of that node. The paths followed by the search are not retained. Although local search algorithms are not systematic, they have two key advantages: (1) they use very little memory, usually a constant amount (2) they can often find reasonable solutions in large or infinite (continuous) state spaces for which systematic algorithms are unsuitable. Local search algorithms are also useful for solving pure optimization problems, in which the aim is to find the best state according to an objective function.

2. Explain the major steps of A* search. (3)

A* Algorithm Steps

1. **Firstly, add the beginning node to the open list**
2. **Then repeat the following step**

– In the open list, find the square with the lowest F cost – and this denotes the current square.
– Now we move to the closed square.
– Consider 8 squares adjacent to the current square and

- Ignore it if it is on the closed list, or if it is not workable. Do the following if it is workable
- Check if it is on the open list; if not, add it. You need to make the current square as this square's a parent. You will now record the different costs of the square like the F, G and H costs.
- If it is on the open list, use G cost to measure the better path. Lower the G cost, the better the path. If this path is better, make the current square as the parent square. Now you need to recalculate the other scores – the G and F scores of this square.

– **You'll stop:**

- Check if it is on the open list; if not, add it. You need to make the current square as this square's a parent. You will now record the different costs of the square like the F, G and H costs.
- If it is on the open list, use G cost to measure the better path. Lower the G cost, the better the path. If this path is better, make the current square as the parent square. Now you need to recalculate the other scores – the G and F scores of this square.

– **You'll stop:**

- If you find the path, you need to check the closed list and add the target square to it.
- There is no path if the open list is empty and you could not find the target square.

3. **Now you can save the path and work backwards starting from the target square, going to the parent square from each square you go, till it takes you to the starting square. You've found your path now.**

Why is A* Search Algorithm Preferred?

3. Write a brief essay on heuristic functions. (3)

Ans: The heuristic function is a way to inform the search about the direction to a goal. It provides an informed way to guess which neighbor of a node will lead to a goal.

d) Heuristic Functions

Heuristics is often expressed as a function of a state:

h(s) = Estimated cost of the cheapest path from s to a goal state.

Examples:

i) 8-puzzle problem

Select the generated state s with

$h_1(s)$ = number of tiles not matching those in the goal.

6	3	4
7		5
2	1	8

Initial

	1	2
3	4	5
6	7	8

Goal

h_1(Initial) = 6

Finding a good heuristic function is a big problem. Statistical and probabilistic methods help in that aspect. And the good heuristic functions are said to be admissible and consistent.

4. How evolution is modeled in Genetic algorithms? (2)

Ans: Evolution is modeled in Genetic algorithms based on population genetics and survival of the fittest Here, parent states are chosen from a set after rating with an objective function, which is called fitness function. Also, ‘Crossover’ and ‘Mutation’ are accommodated here.

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1. Distinguish between blind and informed search. (2) Ans:

Blind Search	Informed Search

It doesn’t use knowledge for the searching process.	It uses knowledge for the searching process.
It finds solutions more quickly.	It finds solutions more quickly.
It is always complete.	It may or may not be complete.
Cost is high.	Cost is low.
It consumes moderate time.	It consumes less time.
No suggestion is given regarding the solution in it.	It provides the direction regarding the solution.
It is more lengthy while implementation.	It is less lengthy while implementation.
Depth First Search, Breadth First Search	Greedy Search, A* Search, Graph Search

2. Describe the major steps of a typical Hill-climbing local search algorithm. (3) Ans:

Typical Hill climbing is the simplest way to implement a hill climbing algorithm. It only evaluates the neighbor node state at a time and selects the first one which optimizes current cost and sets it as a current state. It only checks it's one successor state, and if it finds better than the current state, then move else be in the same state.

The major steps of a typical Hill -climbing local search algorithm -

1. Evaluate the initial state, if it is a goal state then return success and Stop.
2. Loop Until a solution is found or there is no new operator left to apply.
3. Select and apply an operator to the current state.
4. Check new state:
   a. If it is a goal state, then return to success and quit.
   b. Else if it is better than the current state then assign a new state as a current state.
   c. Else if not better than the current state, then return to step2.
5. Terminate/exit.

3. How reproduction is modeled in Genetic algorithms? (2) Ans:

The reproduction function determines how to expand the population based on the existing members. Reproduction is modeled in three steps in Genetic Algorithm. These are-

1. **Mutation**: The simplest reproduction method is mutation, where each new member is based on a single individual
2. **Crossover:** Crossover is a more complex method of reproduction, where each new

individual is based on some combination of existing individuals.
The reproduction function returns the new population, which may be the same size or a different size than the original population.

4. Illustrate the process of updating the Possible Path in Greedy Best-first search. (3) Ans:

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1. **How do you explain a Goal test in search problems? (2)**

Ans: The goal test, which determines whether a given state is a goal state. Sometimes there is an explicit set of possible goal states, and the test simply checks whether the given state is one of them.Sometimes the goal is specified by an abstract property rather than an explicitly enumerated set of states. For example, in chess, the goal is to reach a state called "checkmate," where the opponent's king is under attack and can't escape.

2. **Explain three important decisions about A* search. (3) Ans:**

c) Important decisions about A* search:

1. A* in tree search is optimal if h(n) is admissible.
2. A* in general graph search is optimal if h(n) is consistent.
3. If h(n) is consistent, then the values of f(n) along any path are non decreasing.
4. A* expands all nodes with f(n) < C*, where C* is the cost of the optimal solution.
5. A* is complete, assuming that there are only finite number of nodes with cost less than or equal to C*.
6. A* is said to be optimally efficient, that is, no other optimal algorithm is guaranteed to expand fewer nodes than A* does.

3. **Write a brief essay on 'Some real problems require Hill-climbing local search'. (3)**

Ans: The hill-climbing algorithm can be applied in the following areas:
Marketing: A hill-climbing algorithm can help a marketing manager to develop the best marketing plans. This algorithm is widely used in solving Traveling-Salesman problems. It can help by optimizing the distance covered and improving the travel time of sales team members. The algorithm helps establish the local minima efficiently.

Robotics:

Hill climbing is useful in the effective operation of robotics. It enhances the coordination of different systems and components in robots.

Job Scheduling:

The hill climbing algorithm has also been applied in job scheduling. This is a process in which system resources are allocated to different tasks within a computer system. Job scheduling is achieved through the migration of jobs from one node to a neighboring node. A hill-climbing technique helps establish the right migration route.

3.4. Hill-climbing Local Search and Optimization Strategy

A) Ideas Behind

Specific real-world problems can be explained with the help of the 8-queens problem:

- Integrated circuit designing
- Telecommunications network optimization
- Automatic programming
- Factory-floor layout designing
- Job-shop scheduling
- Vehicle routing
- Portfolio management

...

4. Elaborate the concept of Mutation in connection with Genetic algorithms. (2) Ans:

6) Each member of new generation can undergo, with a small independent probability, the process of 'mutation'.

Mutation:
Say, [e. 25356321] → [e′: 2**6**356321].

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1. Write a short note on 'Optimal solution of a search problem'. (2)

9. **Optimal Solution**
Solution that has the lowest path cost

A specific path, P_s found in the directed graph.

2. Illustrate the odds of the environment of Hill-climbing local search problems. (3)

ii) Measures to face the odds of the 'landscape' (environment)

a. Sideways move: Try to go out of a 'shoulder' by insisting on a particular 'direction'.

b. Random restart hill climbing: If stuck up at a local maximum, then begin with a new randomly generated state; Not bad if not so many local maxima out there;

c) Stochastic Hill-climbing: Choose one at random from among the uphill moves.

d) First-choice Hill-climbing: Choose the 1st randomly generated successor that is better than the current state.

3. What do you know about the evaluation function of Greedy best-first search? (2)

3. A node is selected for expansion based on an evaluation function, f(n), which is taken
$f(n) = h(n)$,
that means, expand the node which 'is closest to the goal according to the heuristic function' or 'seems best'.

4. Narrate the ideas that stand behind Genetic algorithms. (3)

3.5. Genetic Algorithms

A) Ideas Behind

1) A GA is a variant of stochastic beam search:

- local search
- more than one current states
- probabilistic in choosing the successor state

2) A GA models reproduction involving two parents:

a successor is generated by combining two states

3) A GA models evolution:

- population genetics
- survival of the fittest

✓Parent states are chosen from a set after rating with an objective function.
✓The function is called fitness function .
✓ 'Crossover' and 'Mutation' are also accommodated.

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1. Describe the conditions under which heuristic functions are considered good. (3)

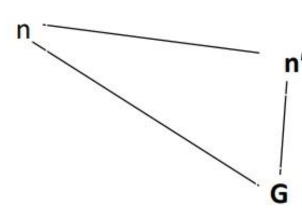
Ans:

i) **Admissibility of heuristic functions**: An admissible heuristic function never over estimates the cost to reach the goal:
h(s) <= Actual cost to reach the goal from s.
h1 and h2 seen above are admissible:
h1 – any tile that is out of place, must be moved at least once h2 – straight line is the shortest of any other lines

g) Monotonicity or Consistency of heuristic functions

$h(n) \le c(n, a, n') + h(n')$, for any node n and every successor n′ of n

It is from triangular inequality: one side is shorter than the sum of the other two.



❖ h_1 and h_2 seen above are also consistent.

❖ Every consistent heuristic function is also admissible, but vice-versa is not always true.

2. **Explain three of the most important features of Hill-climbing Local Search. (3) Ans:**

- **Generate and Test variant:** Hill Climbing is the variant of Generate and Test method. The Generate and Test method produce feedback which helps to decide which direction to move in the search space.
- **Greedy approach:** Hill-climbing algorithm search moves in the direction which optimizes the cost.. It is simply a loop that continually moves in the direction of increasing value—that is, uphill. I
- **No backtracking:** It does not backtrack the search space, as it does not remember the previous states.. Hill climbing does not look ahead beyond the immediate neighbors of the current state.

3. **How the new generation is formed in Genetic algorithms? (2)**

Ans: In genetic algorithms, parents are chosen randomly for ‘crossover’ and a new generation is obtained consisting of the ‘offsprings’. Suppose we have 4 individuals rated using a fitness function where F(a) =23, F(b)=13, F(c)=22.Now ,if 15 is the threshold value parent generation is {a,c,d}.Suppose a=25348716 d=31456321.Now if a and d are chosen for crossover and 4 is the crossover point then .253|48716 314|56321 new generation:e’=25356321 and f=31448716.This is how a new generation is formed in genetic algorithms.

4. **Mark the negative points in Greedy Best-First search. (2) Ans:**

- The strategy is greedy,may not return a optimal solution
- The strategy is not complete in the sense that it may unreasonably lead to great depth.

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1. **Explain the necessity of heuristic functions for problem solving. (2) Ans:**

T3.1.pdf - Google Drive

T3.1.pdf

▪ Informed or Heuristic search helps

- Additional prior knowledge
- Estimation or problem specific rule
- Helps sometimes, but not always
- Heuristics: ‘to find’, ‘to discover’, ‘rule of thumb’ or ‘judgmental technique’

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2. Narrate those major steps of a typical genetic algorithm that are repeatedly executed. (3)

Ans:

T3.5.pdf

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B) Major steps of a typical Genetic algorithm:

Toy example: 8-queens problem

1) Begins with an initial population, that is, set of data structures.

Suppose 4 states: a) 25348716 b) 81765432 c) 42561862 d) 31456321

2) Each individual is rated using a fitness function.

F(a) = 28-h(a) = 28-5 =23, F(b) = 13, F(c) = 22, F(d) = 17.

3) Process terminates if one individual is ‘sufficiently fit’.

Say, the value to be ‘sufficiently fit’ is 25.

4) A parent generation is formed taking better performing individuals.

Parent generation:
Taking 15 as the threshold value, for example, we get {a, c, d}.

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5) Parents are chosen randomly for ‘crossover’, and a new generation is obtained consisting of the ‘offsprings’.

Say, a and d are chosen for crossover, and 4 is the crossover point.
253|**48716** New generation: e. 253**56321**
314|**56321** f. 314**48716**

6) Each member of new generation can undergo, with a small independent probability, the process of ‘mutation’.

Mutation:
Say, [e. 25356321] → [e’: **2**6356321].

7) Parent generation and new generation form new population for running the next cycle beginning from step 2.

New population:
{a, c, d, e’, f}

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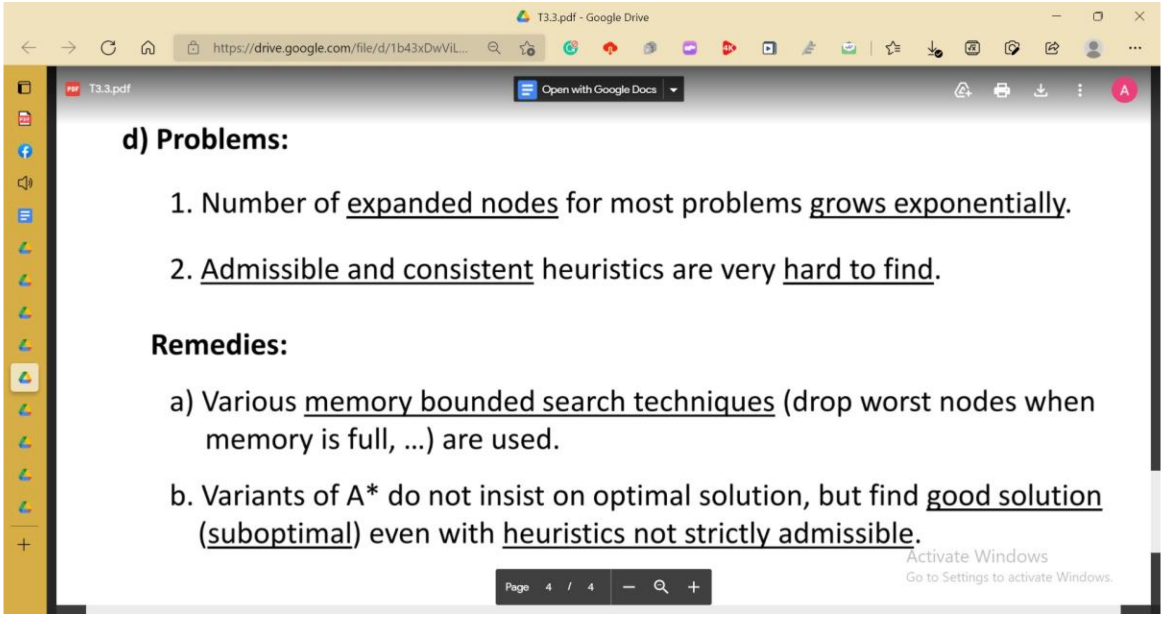
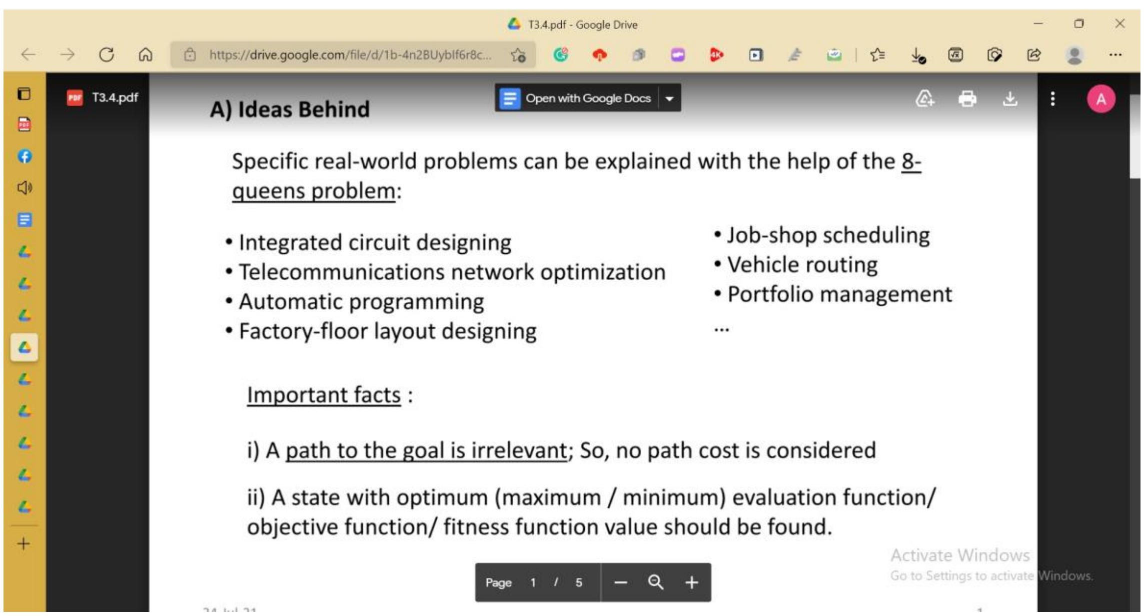
3. What are the problems with A* search, and how those can be addressed? (3) Ans:



4. Write a short note on Random restart Hill-climbing search. (2) Ans:



1. Write a short note on ‘Local maximum’. (2)

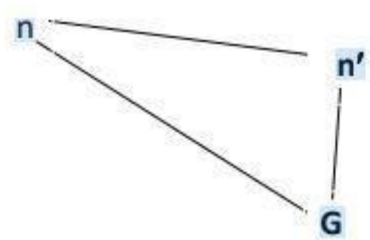
Local maximum: It is a state which is better than its neighboring state however there exists a state which is better than it(global maximum). This state is better because here the value of the objective function is higher than its neighbors.

2. Narrate the major steps of A* search that is repeatedly executed. (3)

The most widely known form of best-first search is called A∗ search. It evaluates nodes by combining g(n), the cost to reach the node, and h(n), the cost to get from the node to the goal: f(n) = g(n) + h(n). Since g(n) gives the path cost from the start node to node n, and h(n) is the estimated cost of the cheapest path from n to the goal, we have f(n) = estimated cost of the cheapest solution through n.
Thus, if we are trying to find the cheapest solution, a reasonable thing to try first is the node with the lowest value of g(n) + h(n). It turns out that this strategy is more than just reasonable: provided that the heuristic function h(n) satisfies certain conditions, A∗ search is both complete and optimal. The algorithm is identical to UNIFORM-COST-SEARCH except that A∗ uses g + h instead of g.

3. Write a brief essay on Consistency of heuristic functions. (3)

h(n) c(n, a, n) + h(n), for any node n and every successor n of n



It is from triangular inequality:
one side is shorter than the sum of the ot[illegible]

- ❖ h1 and h2 are consistent.

Where, h1 – any tile that is out of place, must be moved at least once h2 – straight line is the shortest of any other lines

❖ Every consistent heuristic function is also admissible, but vice-versa is not always true.

[May work very well on n, while not so well on n.]

4. How Crossover is executed in Genetic algorithms? (2)

1) A Genetic Algorithm is a variant of stochastic beam search.

One of the major steps of a typical Genetic algorithm is "Crossover" where parents are chosen randomly for the crossover and the new generation is obtained consisting of the "offspring".

Say, a and d are chosen for crossover, and 4 is the crossover point.

253|**48716**	New generation:	e. 253**56321**
314|**56321**		f. 314**48716**

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1. Write a short note on 'Admissibility of heuristic functions'. (2)

In computer science, specifically in algorithms related to pathfinding, a heuristic function is said to be **admissible** if it never overestimates the cost of reaching the goal, i.e. the cost it estimates to reach the goal is not higher than the lowest possible cost from the current point in the path.

6	3	4
7		5
2	1	8

Initial

	1	2
3	4	5
6	7	8

Goal

For 8-puzzle problem

Select the generated state s with

h1(s) = number of tiles not matching those in the goal.

h1 is admissible:

h1 – any tile that is out of place, must be moved at least once

2. **Describe the repeatedly executed major steps of a typical Hill-climbing local search. (3)**
3. **How randomness is applied in Genetic algorithms? (2)**

The hill-climbing search algorithm, which is the most basic local search technique. At each step the current node is replaced by the best neighbor; in this version, that means the neighbor with the highest VALUE, but if a heuristic cost estimate h is used, we would find the neighbor with the lowest h.

Algorithm: 1. Uses a single current state;

2. Investigates neighbors generated by the successor function;

3.Terminates if a state with optimal / 'sufficient' value found;

```
function HILL-CLIMBING(problem) returns a state that is a local maximum
    current ← MAKE-NODE(problem.INITIAL-STATE)
    loop do
        neighbor ← a highest-valued successor of current
        if neighbor.VALUE ≤ current.VALUE then
            return current.STATE
        current ← neighbor
```

In Generic algorithm, the evolution usually starts from a population of randomly generated individuals and happens in generations.

In Selection, accumulated normalized fitness values are computed: the accumulated fitness value of an individual is the sum of its own fitness value plus the fitness values of all the previous individuals. The selected individual is the first one whose accumulated normalized value is greater than or equal to *R*. Here, R is a random number chosen between 0 and 1.

In Crossover, parents are chosen randomly, and a new generation is obtained consisting of the 'offspring'. A common method of implementing the mutation operator involves generating a random variable for each bit in a sequence. This random variable tells whether a particular bit will be flipped.

4. Explain the positive and negative sides of Greedy Best-first search. (3)

Positive:

1. A node is selected for expansion based on an evaluation function, f(n), which is taken f(n) = h(n), that means, expand the node which 'is closest to the goal according to the heuristic function' or 'seems best'.
2. A Priority Queue (PQ), which contains nodes in ascending order of h-values, is also maintained. The PQ offers the node for expansion.
3. A Possible Path (PP) is maintained that contains nodes currently supposed to be in the solution.
4. A tree of visited nodes along with their children is also maintained which helps to update PQ and PP.
5. The process begins by placing the source node (initial state) in the empty PQ, and initiating a tree by placing that node as its root.
   The process terminates when the destination node (goal state) is placed in the PQ, and selected, consequently.

Negative:

1. The strategy is 'greedy', may return a non-optimal solution.
2. The strategy is also not complete in the sense that it may unreasonably lead to great depth.

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1. Write a short note on 'Solution to AI problems'. (2) Ans:

2. Why and how a tree of nodes is maintained in Greedy Best-first search? (3)

Ans: A tree of visited nodes along with their children is maintained in Greedy Best-first search which helps to update the Priority Queue (PQ) and the Possible Path (PP). The Priority Queue contains nodes in ascending order of h-values. The PQ offers the node for expansion. The Possible Path (PP) contains nodes currently supposed to be in the solution.

<u>How a tree of nodes is maintained:</u>

- The process begins by placing the source node (initial state) in the empty PQ, and initiating a tree by placing that node as its root.
- The process terminates when the destination node (goal state) is placed in the PQ, and selected, consequently.
- The 1st node from the PQ is selected repeatedly, and each time the tree, the PQ and the PP are updated:

✓ The node in the tree is marked visited and its neighbors from the graph are added to the tree as its children, while no repeated node is allowed in the tree;

✓ The node itself is deleted from the PQ, but its children added to the tree are also added to the PQ.

✓ The PP is straightened up to the root from the selected node.

3. For what type of problems Hill-climbing local search is appropriate? (2)

Ans: Hill Climbing local search is appropriate for mathematical optimization problems in the field of Artificial Intelligence. Mathematical optimization problems imply that hill-climbing solves the problems where we need to maximize or minimize a given real function by choosing values from the given inputs. Hill Climbing technique can be used to solve many problems, where the current state allows for an accurate evaluation function, such as Network-Flow, Traveling Salesman problem, 8-Queens problem, Integrated Circuit design, etc.
Given a large set of inputs and a good heuristic function, it tries to find a sufficiently good solution to the problem. This solution may not be the global optimal maximum.

4. Elaborate the concept of Population in connection with Genetic algorithms. (3)

Ans: Genetic Algorithms begin with a set of randomly generated states, called the population. Each state, or individual, is represented as a string over a finite alphabet—most commonly, a string of 0s and 1s.The population has a fixed size. As new generations are formed, individuals with least fitness die, providing space for new offspring. The sequence of phases is repeated to produce individuals in each new generation which are better than the previous generation.
The production of the next generation of states is rated by the objective function, or (in GA terminology) the fitness function. A fitness function should return higher values for better states. A parent generation is formed taking better performing individuals. Parents

are chosen randomly for 'crossover', and a new generation is obtained consisting of the 'offsprings'. Each member of the new generation can undergo, with a small independent probability, the process of 'mutation'. Parent generation and new generation form a new population for running the next cycle.

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1. **Write a short note on 'Path cost of AI problems'. (2)**

Ans: Path-finding is the task of finding the shortest path between two given nodes in a graph, and has been studied in computer science (CS) for almost forty years. Theoretical advances are still being made and real-world applications abound. For example, there are commercial products that find routes optimizing the cost or time to drive between two locations in a city or country. Path-finding arises as a subtask in many areas of artificial intelligence (AI). For example, in natural language understanding, lexical disambiguation has been partially solved by finding paths between word senses in a knowledge base. A path cost function assigns a numeric cost to each path. One that has the lowest path cost among the solutions gives the desired resulting condition in a given problem, and what search algorithms are looking for.

2. **Illustrate the concept of local maxima in connection with Hill-climbing local search. (3)**

Ans: A local maximum is a peak that is higher than each of its neighboring states but lower than the global maximum. Hill-climbing algorithms that reach the vicinity of a local maximum will be drawn upward toward the peak but will then be stuck with nowhere else to go. Figure 4.1 illustrates the problem schematically.

To illustrate hill climbing, we will use the 8-queens problem .Local search algorithms typically use a complete-state formulation, where each state has 8 queens on the board, one per column. The successors of a state are all possible states generated by moving a single queen to another square in the same column (so each state has 8 × 7 = 56 successors). The heuristic cost function h is the number of pairs of queens that are attacking each other, either directly or indirectly. The global minimum of this function is zero, which occurs only at perfect solutions.

In each case, the algorithm reaches a point at which no progress is being made. Starting from a randomly generated 8-queens state, steepest-ascent hill climbing gets stuck 86% of the time, solving only 14% of problem instances. It works quickly, taking just 4 steps on average when it succeeds and 3 when it gets stuck - not bad for a state space with 88 ≈ 17 million states.

The success of hill climbing depends very much on the shape of the state-space landscape: if there are few local maxima and plateaus, random-restart hill climbing will find a good solution very quickly.

3. What do you know about the evaluation function of A* search? (2)

Ans: A* algorithm is a form of heuristic search that tries to find the cheapest path from the initial state to the goal. Its characteristic feature is the evaluation function. This is the sum of two components: the estimated minimum cost of a path from the initial state to the current state, and the estimated cost from the current state to the goal. The first component can be calculated if the search space is a tree, or it can be approximated by the cheapest known path if the search space is a graph. The second component must be defined, like any evaluation function, with respect to the domain. The heuristic power of this method depends on the properties of the evaluation function.
It evaluates nodes by combining g(n), the cost to reach the node, and h(n.), the cost to get from the node to the goal: f(n) = g(n) + h(n). Since g(n) gives the path cost from the start node to node n, and h(n) is the estimated cost of the cheapest path from n to the goal.

4. Narrate the importance of Crossover and Mutation used in Genetic algorithms. (3)

Ans: A genetic algorithm is a stochastic hill-climbing search in which a large population of states are maintained. New states are generated by mutation and by crossover, which combines pairs of states from the population.
Crossover and mutation perform two different roles. Crossover (like selection) is a convergence operation which is intended to pull the population towards a local minimum/maximum. In an interesting aside, crossover is not one of the original genetic operators; Holland's original thesis used only selection and probabilistic mutation.
Mutation is a divergence operation. It is intended to occasionally break one or more members of a population out of a local minimum/maximum space and potentially discover a better minimum/maximum space.
Since the end goal is to bring the population to convergence, selection/crossover happen more frequently (typically every generation). Mutation, being a divergence operation, should happen less frequently, and typically only affects a few members of a population (if any) in any given generation.

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1. Illustrate the conditions under which heuristic functions are considered good. (3)

Ans:

e) Good heuristic functions

- Finding a good heuristic function is a big problem.
- Statistical and probabilistic methods help.
- May require additional computational cost (as for h_1), or prior knowledge (as for h_2).
- Good heuristic functions are said to be admissible or consistent.

f) Admissibility of heuristic functions

❑ An admissible heuristic function never over estimates the cost to reach the goal:

h(s) ≤ Actual cost to reach the goal from s

❑ h_1 and h_2 seen above are admissible:

h_1 – any tile that is out of place, must be moved at least once

h_2 – straight line is the shortest of any other lines

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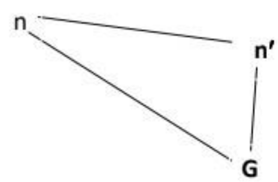
g) Monotonicity or Consistency of heuristic functions

$h(n) \leq c(n, a, n') + h(n')$, for any node n and every successor n′ of n

It is from triangular inequality: one side is shorter than the sum of the other two.



❖ h_1 and h_2 seen above are also consistent.

❖ Every consistent heuristic function is also admissible, but vice-versa is not always true.

[May work very well on n′, while not so well on n.]

2. Justify the naming of the strategy called Hill-climbing Local Search. (3)

Ans: Hill climbing algorithm is a local search algorithm which continuously moves in the direction of increasing elevation/value to find the peak of the mountain or best solution to the problem. It terminates when it reaches a peak value where no neighbor has a higher value.It is also called greedy local search as it only looks to its good immediate neighbor state and not beyond that. It is an iterative algorithm that starts with an arbitrary solution to a problem, then attempts to find a better solution by making an incremental change to the solution.

3. How the new generation is formed in Genetic algorithms? (2) Ans:

B) Major steps of a typical Genetic algorithm:

Toy example: 8-queens problem

1) Begins with an initial population, that is, set of data structures.

   Suppose 4 states: a) 25348716 b) 81765432
   c) 42561862 d) 31456321

2) Each individual is rated using a fitness function.

   F(a) = 28-h(a) = 28-5 =23, F(b) = 13, F(c) = 22, F(d) = 17.

3) Process terminates if one individual is 'sufficiently fit'.

   Say, the value to be 'sufficiently fit' is 25.

4) A parent generation is formed taking better performing individuals.

   Parent generation:
   Taking 15 as the threshold value, for example, we get {a, c, d}.

5) Parents are chosen randomly for 'crossover', and a new generation is obtained consisting of the 'offsprings'.

   Say, a and d are chosen for crossover, and 4 is the crossover point.
   253|**48716** New generation: e. 253**56321**
   314|**56321** f. 314**48716**

6) Each member of new generation can undergo, with a small independent probability, the process of 'mutation'.

   Mutation:
   Say, [e. 25356321] → [e': 2**6**356321].

7) Parent generation and new generation form new population for running the next cycle beginning from step 2.

   New population:
   {a, c, d, e', f}

4. Explain the negative side of A* search. (2) Ans:

1. Number of expanded nodes for most problems grows exponentially.

2. Admissible and consistent heuristics are very hard to find.

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1. **Illustrate the concept of consistency of heuristic functions with a concise example. (3)**
Ans: 110 has filled it already. <3

2. **Why is a genetic algorithm called a local search? (2)**
Ans: Genetic Algorithms have been seen as search procedures that can quickly locate high performance regions of vast and complex search spaces, but they are not well suited for fine-tuning solutions, which are very close to optimal ones. However, genetic algorithms may be specifically designed to provide an effective local search as well. In fact, several genetic algorithm models have recently been presented with this aim.genetic al- gorithms may be specifically designed to provide an effective local search as well.

3. **Explain the positive and negative sides of Greedy Best-first search. (3) Ans:** 121 has filled it already. <3

4. **Write a short note on 'Stochastic Hill-climbing search'. (2)**
Ans: Stochastic hill climbing is a variant of the basic hill climbing method. While basic hill climbing always chooses the steepest uphill move, "stochastic hill climbing chooses at random from among the uphill moves; the probability of selection can vary with the steepness of the uphill move**.**
It is also a local search algorithm, meaning that it modifies a single solution and searches the relatively local area of the search space until the local optima is located. This means that it is appropriate on unimodal optimization problems or for use after the application of a global optimization algorithm**.**

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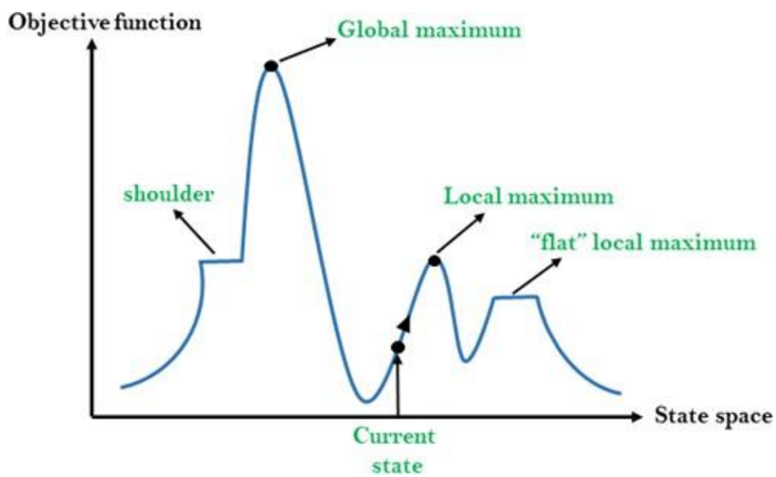
1. **Write a short note on 'Hill-climbing search'. (2)**

Ans: Hill climbing algorithm is a local search algorithm which continuously moves in the direction of increasing elevation/value to find the peak of the mountain or best solution to the problem. It terminates when it reaches a peak value where no neighbor has a higher value. A node of hill climbing algorithm has two components which are state and value. In this algorithm, we don't need to maintain and handle the search tree or graph as it only keeps a single current state.

Features of Hill Climbing: 1. Generate and Test variant

2. Greedy approach 3.No backtracking



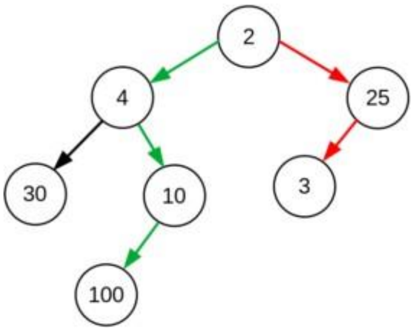
X-axis: denotes the state space i.e. states or configuration our algorithm may reach. Y-axis: denotes the values of objective function corresponding to a particular state.

2. Narrate the major steps of A* search that is repeatedly executed. (3) Ans: See ID-120's 2nd ans.

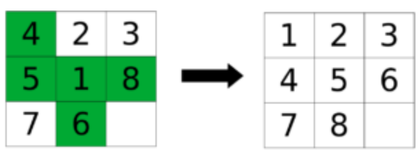
3. Illustrate the concept of heuristic functions. (3)

Ans: A heuristic function (algorithm) or simply a heuristic is a shortcut to solving a problem when there are no exact solutions for it or the time to obtain the solution is too long. The goal of the function is to find a faster solution or an approximate one, even if it is not optimal. In other words, when using a heuristic, we trade accuracy for the speed of the solution.

For example, greedy algorithms usually produce quick but suboptimal solutions. A greedy algorithm to find the largest sum in the following tree would go for the red path while the optimal path is the green one:



Heuristics don’t always lead to a lower cost. However, those that don’t overestimate the true or the lowest possible cost of a solution are called admissible heuristics.



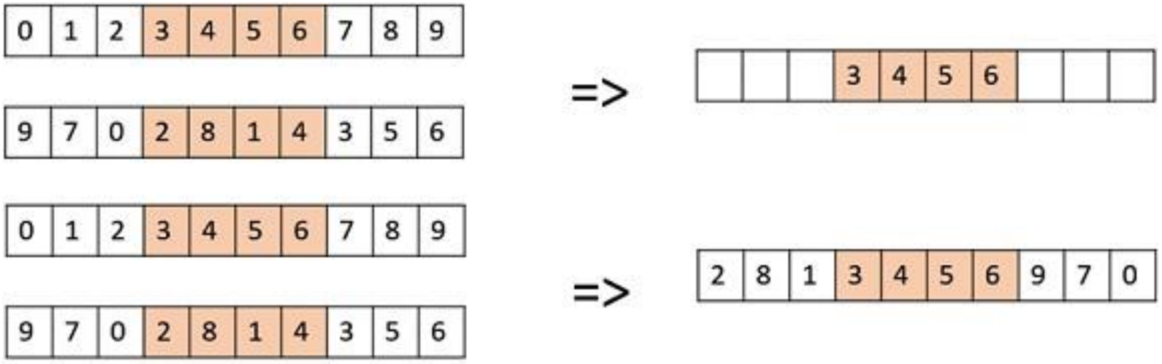
Here we start from the state on the left, and we want to reach the goal state on the right. As a heuristic solution to this puzzle, we can think of the Hamming distance between the two states, which is the number of misplaced tiles highlighted in green. Here, we have relaxed the constraints of the problem by assuming that we can just pick up a tile and put it in its right place.

There are numerous problems for which we can come up with a heuristic algorithm to solve.Few of them are :

1. A* Search
2. Traveling Salesman Problem (TSP)
3. Distance on Grid Maps

4. **How Crossover is executed in Genetic algorithms? (2)**

Ans: Create two random crossover points in the parent and copy the segment between them from the first parent to the first offspring. Now, starting from the second crossover point in the second parent, copy the remaining unused numbers from the second parent to the first child, wrapping around the list.



Repeat the same procedure to get the second child

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1. Distinguish between blind and informed search. (2)

Blind or Uniformed Search

It is a search without "information" about the goal node.

An example an is breadth-first search (BFS). In BFS, the search proceeds one layer after the other. In other words, nodes in the same layer are first visited before nodes in successive layers. This is performed until a node that is "expanded" is the goal node. In this case, no information about the goal node is used to visit, expand or generate nodes.

We can think of a blind or uniformed search as a brute-force search.

Heuristic or Informed Search

It is a search with "information" about the goal.

An example of such type of algorithm is A*. In this algorithm, nodes are visited and expanded using also information about the goal node. The information about the goal node is given by an **heuristic function** (which is a function that associates information about the goal node to each of the nodes of the state space). In the case of A*, the heuristic information associated with each node `n` is an estimate of the distance from `n` to the goal node.

Ans: We can think of a informed search as a approximately "guided" search.

2. Describe the repeatedly executed major steps of a typical Genetic algorithm. (3)

B) Major steps of a typical Genetic algorithm:

Toy example: 8-queens problem

1) Begins with an initial population, that is, set of data structures.

Suppose 4 states: a) 25348716 b) 81765432
c) 42561862 d) 31456321

2) Each individual is rated using a fitness function.

F(a) = 28-h(a) = 28-5 =23, F(b) = 13, F(c) = 22, F(d) = 17.

3) Process terminates if one individual is 'sufficiently fit'.

Say, the value to be 'sufficiently fit' is 25.

4) A parent generation is formed taking better performing individuals.

Parent generation:
Taking 15 as the threshold value, for example, we get {a, c, d}.

Ans: 2/25/2022 2

3. Explain the positive side of Hill-climbing local search. (2) Ans: See ID-106's 3rd ans.

4. Illustrate the process of updating the Possible Path in A* search. (3) Ans: do it yourself